

Bimsara Madhawa Udurawana

Computer Science & Engineering Undergraduate | Data Science & Engineering

bimsarau.23@cse.mrt.ac.lk | bimsaraudurawana@gmail.com | +94 70 337 1001 | Katugastota, Sri Lanka
GitHub: <https://github.com/BimsaraU>

PROFILE

Computer Science & Engineering undergraduate at the University of Moratuwa (CGPA 3.73/4.00), specialising in Data Science & Engineering, with a focus on computer vision and time-series analysis. Experienced in open-vocabulary and zero-shot object detection, embedding-based semantic retrieval, forecasting from noisy real-world signals, and end-to-end MLOps pipelines spanning experiment tracking, automated retraining and edge deployment. Complements this with FPGA hardware acceleration and RTL design verification, delivering INT8 quantised CNN inference accelerators verified bit-exact on real silicon.

EDUCATION

B.Sc. (Hons) in Engineering — Computer Science & Engineering

Specialisation: Data Science & Engineering

2024 – Present

University of Moratuwa, Sri Lanka | **Current CGPA: 3.73 / 4.00** | **Semester 4 GPA: 3.70 / 4.00** | Semester GPAs: Sem 1 — 4.00, Sem 2 — 3.59, Sem 3 — 3.71, Sem 4 — 3.70

Relevant coursework: Computer Architecture, Digital System Design, Data Structures & Algorithms, Machine Learning, Database Systems, Operating Systems, Data Science.

G.C.E. Advanced Level — Physical Science Stream, St. Anthony's College, Kandy

2023

3 A passes: Combined Mathematics, Physics, Chemistry | A pass for General English | G.C.E. Ordinary Level: 9 A passes.

TECHNICAL SKILLS

Computer Vision: PyTorch, TensorFlow, OpenCV, Ultralytics YOLO, Object Detection & Segmentation, Open-Vocabulary & Zero-Shot Detection, Image Preprocessing & Augmentation, Embedding Models & Semantic Search, Knowledge Distillation, Transfer Learning, Model Evaluation & Benchmarking.

Time-Series & Data Science: Time-Series Forecasting, Cross-Correlation & Lag Analysis, Feature Engineering, Seasonality & Trend Decomposition, Anomaly Detection, Regression Modelling, Statistical Analysis, scikit-learn, Pandas, NumPy, Apache Spark, SQL.

MLOps & Model Optimisation: MLflow, Weights & Biases, DVC, ONNX Runtime, Model Quantisation & Pruning, Experiment Tracking & Model Registry, Automated Retraining Pipelines, Streaming Data Pipelines (Kafka, MQTT), Edge Deployment.

DevOps & Cloud Infrastructure: Docker, Kubernetes, Terraform (IaC), GitOps (ArgoCD), CI/CD (GitHub Actions), Grafana, Prometheus, Observability & Monitoring, API Gateways, Git, Linux.

Hardware Design & RTL: SystemVerilog, VHDL, Verilog, RTL Micro-architecture, Systolic Arrays, Pipelining, FSM Design, AXI4 Interconnect, DMA Engines, Network-on-Chip, BRAM & DSP Mapping, Timing Closure, Resource & Power Optimisation, RISC-V, SoC Integration, Embedded Interfaces.

Design Verification: Self-checking Testbenches, Directed & Constrained-Random Verification, UVM, Assertion-Based Verification (SVA), Functional Coverage, Golden-Model Co-Simulation, Regression Automation, cocotb, Hardware-in-the-Loop Bring-up.

EDA & Toolchains: AMD/Xilinx Vivado Design Suite, QuestaSim, Vitis, Vivado HLS, Synthesis & Implementation Flows, TCL Scripting, Xilinx Artix-7 / Kintex-7 / UltraScale+ FPGAs.

Programming & Software: Python, C++, C, Java, JavaScript, TypeScript, SQL, Bash; React, Next.js, Three.js, Vite, Spring Boot, Node.js, Express.js, TailwindCSS, PostgreSQL, InfluxDB, Redis, MongoDB, REST APIs, WebSockets, Microservices.

Digital Media Design: Adobe Photoshop, Illustrator, InDesign, Lightroom, After Effects, Premiere Pro, Figma, Canva, Meta Business Suite.

Languages: English (Fluent), Sinhala (Native).

PROJECTS

Edge-Optimized Task-Driven Detection: A VEGA-FPGA Co-Design Using Semantic Pre-Filtering and YOLO26n

Hardware/software co-designed zero-shot detection SoC on a Digilent Genesys 2 (Kintex-7 XC7K325T-2) with a C-DAC VEGA RV32 RISC-V host, entered in the DVCon 2026 accelerator track. Built an INT8 convolution accelerator in SystemVerilog — 16×32 MAC tile (512 MACs packed into 256 DSP48E1 via 2-MAC packing), vector unit for bias/requantisation/activation, and an elaboration-time SiLU ROM — exposed as an AXI4 peripheral (AXI4-Lite control slave, AXI4 burst masters to DDR3) with a descriptor-chained GEMM controller, so VEGA issues one kick per inference. Added **semantic pre-filtering**: Model2Vec runs on VEGA in parallel with convolution, cosine-matching prompt embeddings against 80 COCO class embeddings to rank classes and score detections as $conf/(rank+1)$, with an early-exit policy halting on first valid detection; MiniLM (22.7 M params) hit **93.1% top-1 / 100% top-3** intent resolution at **10.3 ms** vs a 195 ms budget, beating MobileCLIP2-B (72.4%, 55.0 ms). Explored a “Sea of PEs” 8×8 array, a Winograd fast-convolution prototype, and a multi-cluster systolic TPU variant (2×2 mesh of 16×16 arrays, **409 GOPS at 200 MHz**, 64-lane vector unit, 2D strided DMA, XY-routed NoC). **Verified on real hardware**: an 8×8 port (64 DSP48E1 PEs, 71% utilisation) under a hard 90-DSP Basys 3 budget was synthesised, placed, routed and run on-board at 100 MHz with **WNS +0.287 ns, bit-exact against a NumPy golden model** across 8/8 randomised trials plus edge cases (248-row MAX_ACTS, full ±128 INT8 range, all—128 asymmetric rail). Authored the BRAM/DSP/im2col verification regression in Vivado xsim, QuestaSim and cocotb. *Technologies*: SystemVerilog, Verilog, RISC-V (C-DAC VEGA), AXI4/AXI4-Lite, Xilinx Vivado, QuestaSim, xsim, cocotb, Kintex-7 & Artix-7 FPGA, DSP48E1, im2col, INT8 Quantisation, Winograd Convolution, Systolic Arrays, NoC, YOLO26n, YOLOE, SentenceTransformers, Model2Vec, MobileCLIP, PyTorch, ONNX, Python, NumPy

Repositories: <https://github.com/TharakaUJ/DVcon-accelerator> | <https://github.com/BimsaraU/DVCon-SittingDucks>

Smart Campus Digital Twin — Cloud-Native IoT & ML Platform

Real-time 3D digital twin of the University of Moratuwa campus, built as a **microservices platform** on Kubernetes. Sensor telemetry flows from MQTT (Mosquitto) through a validating ingest bridge into Apache Kafka, then fans out to a consumer writing time-series to **InfluxDB** (with Flux downsampling tasks) and relational state to PostgreSQL, with Redis caching and Alembic-managed migrations. A FastAPI service exposes REST and **WebSocket** endpoints behind a **Kong** API gateway with **Keycloak** OIDC authentication. The Next.js 16 / React 19 frontend renders an interactive 3D campus in **Three.js** (react-three-fiber) alongside live Recharts dashboards. ML services forecast energy, library and canteen occupancy and detect sensor anomalies, with **MLflow** tracking experiments and a scheduled **retraining daemon** refreshing models automatically. A configurable simulator with anomaly injection drives load testing. Deployed **GitOps-style via ArgoCD** with Kustomize manifests, GitHub Actions CI/CD, and **Prometheus + Grafana** observability dashboards.

Technologies: Python, FastAPI, TypeScript, Next.js, React, Three.js, Kubernetes, Docker, ArgoCD, Kustomize, Apache Kafka, MQTT, InfluxDB, PostgreSQL, Redis, Kong, Keycloak, MLflow, Prometheus, Grafana, GitHub Actions, Alembic

Repository: <https://github.com/Smart-Campus-Digital-Twin/Smart-Campus-Digital-Twin-v4>

TourSL — Sri Lanka Tour Planning Platform

Tour planning and booking platform for multi-day guided tours across Sri Lanka. **Led the frontend:** built the landing experience as a React 18 + TypeScript SPA on Vite with a utility-first Tailwind design system, lucide-react iconography, responsive mobile-first layouts, scroll-driven section reveals and accessible semantic markup; optimised the bundle via code-splitting and shipped continuous preview/production deploys on Vercel. Integrated against the Spring Boot service (layered REST, JWT + Google OAuth, PostgreSQL with Flyway, slot reservation under optimistic locking).

Technologies: React 18, TypeScript, Vite, Tailwind CSS, lucide-react, Responsive UI, Vercel, REST APIs, Java 17, Spring Boot, Spring Security (JWT/OAuth), PostgreSQL, Flyway, Docker

Repositories: <https://github.com/BimsaraU/toursl-landing-page> | <https://github.com/nawarathnaTP/TourSL-Planning-Service>

Live: <https://toursl-landing-page.vercel.app/>

RPAL Interpreter — Functional Language Compiler & Virtual Machine

Complete RPAL interpreter in Java 17 implementing the full pipeline: hand-written lexer, recursive-descent parser producing an AST, an AST standardiser, and a CSE (Control-Stack-Environment) machine flattening the standardised tree into control structures and evaluating them on a stack-based VM with lexical environments. Supports curried multi-argument lambdas, mutual recursion, tuples and escaped strings; CLI switches expose the AST and CSE trace.

Technologies: Java 17, Compiler Design, Lexical Analysis, Recursive-Descent Parsing, AST Transformation, CSE Machine, Virtual Machines, Functional Programming

Repository: <https://github.com/BimsaraU/RPAL-interpreter>

Wallow — Unified Messaging & Social Desktop Client

Windows desktop client consolidating WhatsApp, Telegram, Instagram, TikTok, LinkedIn and Facebook into one lightweight multi-tab window, replacing several heavyweight Electron apps each running its own browser runtime. Built in C++ on the WebView2 runtime with a Dear ImGui immediate-mode interface. Implements active **memory management** — inactive tabs are suspended and their working sets trimmed back to the OS, cutting idle RAM substantially — plus single-instance enforcement via named mutex, system-tray minimisation, per-app cache clearing for privacy, and user-configurable custom tabs, zoom levels, user agents and storage locations. CMake build for Visual Studio 2022.

Technologies: C++, WebView2, Dear ImGui, CMake, Windows API, Visual Studio 2022, Memory Optimisation

Repository: <https://github.com/BimsaraU/Wallow>

Nano Processor Design Project

Progressive FPGA soft-processor in VHDL, iterating from a 4-bit to an 8-bit architecture with Xilinx primitive-level optimisation. Implemented the full datapath and control path from scratch — register file with multiplexed read ports, ripple-carry ALU, program counter with jump/branch logic, instruction ROM, instruction decoder and a hardwired control unit — then extended the ISA beyond the base specification with additional arithmetic and bitwise operations. Verified through waveform simulation against hand-computed expected values and deployed to a Basys 3 board with seven-segment and LED output for live instruction tracing.

Technologies: VHDL, Xilinx Vivado, Basys 3 FPGA, Computer Architecture, RTL Simulation

Repository: <https://github.com/BimsaraU/Nanoprocessor-Design-Project>

Tourism Arrivals Forecasting — Data Science Research

Machine-learning research predicting weekly international tourist arrivals to Sri Lanka from public search-intent signals, testing whether search behaviour leads booking behaviour closely enough to serve as an early demand indicator. Aligned noisy weekly search-volume series against official arrival statistics, ran cross-correlation analysis at varying lags to identify the predictive window, engineered lagged and rolling features, and evaluated regression models against seasonal baselines — giving tourism operators a forecast signal weeks ahead of conventional reporting.

Technologies: Python, scikit-learn, Pandas, NumPy, Time-Series Analysis, Feature Engineering, Statistical Modelling

Repository: <https://github.com/rathishTharusha/DS-research-Tourism>

PhotoTag — GPU-Accelerated Batch Image Processor

Windows desktop application for watermarking and tagging large image batches in a single pass. Detects each image's orientation and adapts watermark placement and grid alignment accordingly, so mixed portrait/landscape sets process correctly without manual intervention, with user-configurable position, scale and opacity controls. Uses DirectX 11 GPU-accelerated rendering for responsive live preview of the full batch, GDI+ for image decoding and encoding, and a Dear ImGui interface for immediate-mode controls.

Technologies: C++, DirectX 11, GDI+, Dear ImGui, Windows API, GPU Rendering

Repository: <https://github.com/BimsaraU/PhotoTag>

SkyNest — Hotel Management System

Full-stack hotel management platform covering the booking lifecycle: room inventory and availability search, reservation creation and modification with date-overlap validation, guest records, check-in/check-out and billing. Normalised PostgreSQL schema with referential integrity and indexed availability queries behind server-side API routes; responsive Next.js + TypeScript frontend with SSR and role-separated admin/guest views. Dockerised, deployed on Vercel.

Technologies: TypeScript, Next.js, React, PostgreSQL, SQL, Docker, REST APIs, Server-Side Rendering, Vercel

Repository: <https://github.com/BimsaraU/SkyNest> | *Live:* <https://sky-nest.vercel.app/>

LEADERSHIP & EXPERIENCE

Director, Digital Media

CSESS, University of Moratuwa (*Mar 2026 – Present*)

Assistant Director, Digital Media

CSESS, University of Moratuwa (*Aug 2025 – Mar 2026*)

Organizing Committee Lead

Cybercon 2026 (*Mar 2026 – Present*)

Assistant Pillar Head, Creative Design

MoraSpirit (*Mar 2025 – Mar 2026*)

Organizing Committee Co-Lead, Design Team

IESL Robogames 2026 (*Dec 2025 – Apr 2026*)

Organizing Committee Co-Lead, Design Committee

Hit The Grounds 2025 (*2025*)

Co-Founder & Head of Design

ZeroToHero (*2023 – Present*)

President, Astronomical Society

St. Anthony's College, Kandy (*2022 – 2023*)

Additional roles: Team Member, CS40 Core Marketing — University of Moratuwa (*Feb 2025 – Present*); Team Member, MarCom IGV & B2B Showcasing MST — AIESEC in Colombo South (*Feb 2025 – Aug 2025*); Organizing Committee Member (Design Team) — MoraForesight, Sakura; Organizing Committee Member (Marketing Team) — SpiritX by MoraSpirit360, Woof Roof, Cricket Fiesta; Pillar Member, Creative Design Pillar — MoraSpirit.

AFFILIATIONS & PERSONAL DETAILS

Affiliations: IEEE Student Branch, University of Moratuwa | CSESS | MoraSpirit | AIESEC in Colombo South | Rotaract Club of University of Moratuwa

Full Name: Kulasekara Mudiyanseelage Hanthanapitiye Bimsara Madhawa Udurawana

Citizenship: Sri Lankan | **Address:** 22/2, Pallethalawinna, Katugastota, Sri Lanka

REFERENCES

Prof. Dulani Meedeniya

Department of Computer Science and Engineering, University of Moratuwa | dulanim@cse.mrt.ac.lk